Question ID: deee9063

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

Line k is defined by $y = 6x + 4$. Line j is parallel to line k in the *xy*-plane and passes through the point $(0, 5)$. Which equation defines line j?

Answer

A. $y = 6x + 5$

B. $y = -5x + 5$

C. $y = -6x + 5$

D. $y = 5x + 5$

Correct Answer: A

Rationale

Choice A is correct. An equation that defines a line in the *xy*-plane can be written as $y = mx + b$, where m is the slope of the line and $(0, b)$ is its *y*-intercept. It's given that line k is defined by the equation $y = 6x + 4$; therefore, the slope of line k is 6. Since line j is parallel to line k in the *xy*-plane and parallel lines have equal slopes, it follows that the slope of line j is 6. It's also given that line j passes through the point $(0, 5)$, which is its *y*-intercept. Substituting 6 for m and 5 for b in the equation $y = mx + b$ yields $y = 6x + 5$. Therefore, the equation $y = 6x + 5$ defines line j.

Choice B is incorrect. This equation defines a line that has a slope of -5, not 6.

Choice C is incorrect. This equation defines a line that has a slope of -6, not 6.

Choice D is incorrect. This equation defines a line that has a slope of 5, not 6.